Crystallization mechanism of amorphous indium-based oxide films

The review of “Temporal Evolution of Microscopic Structure and Functionality during Crystallization of Amorphous Indium-Based Oxide Films”

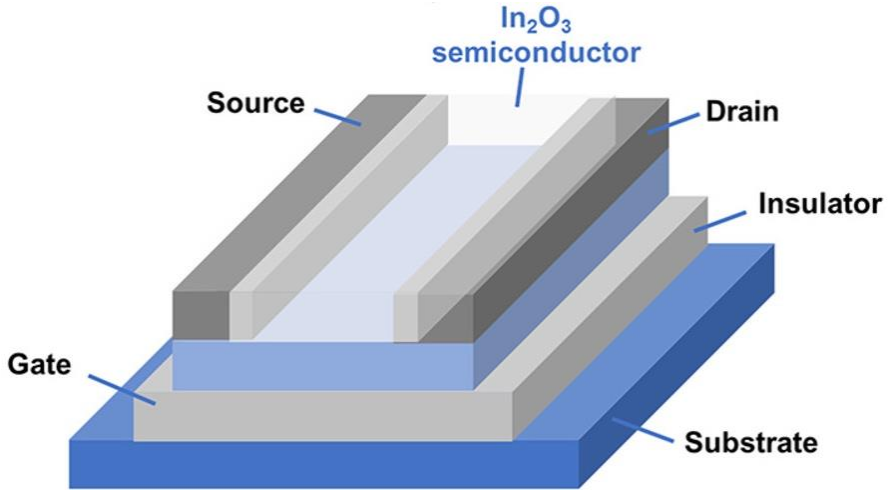
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Introduction

Background about amorphous Indium-based oxide films

Why we choose amorphous Indium-based oxide films?

- Have basic electronic and optical functionalities
- Withstand large strains/stresses or geometrical deformation



Applications of indium-based amorphous oxides

- Amorphous(a-) In_2O_3 and a-ITO → transparent conductive oxide(TCO)
- a-IGO and a-IGZO → semiconductor channel materials in TFT

Amorphous thin films and annealing process

- Low temperature process ⇒ diffusion ↓ ⇒ structural relaxation ↓ ⇒ forming amorphous structures
- Annealing can modulate residual stress and improve the film quality with reduced surface roughness and defects present in the films.

  → The material properties of amorphous films are improved by the annealing process.

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Introduction

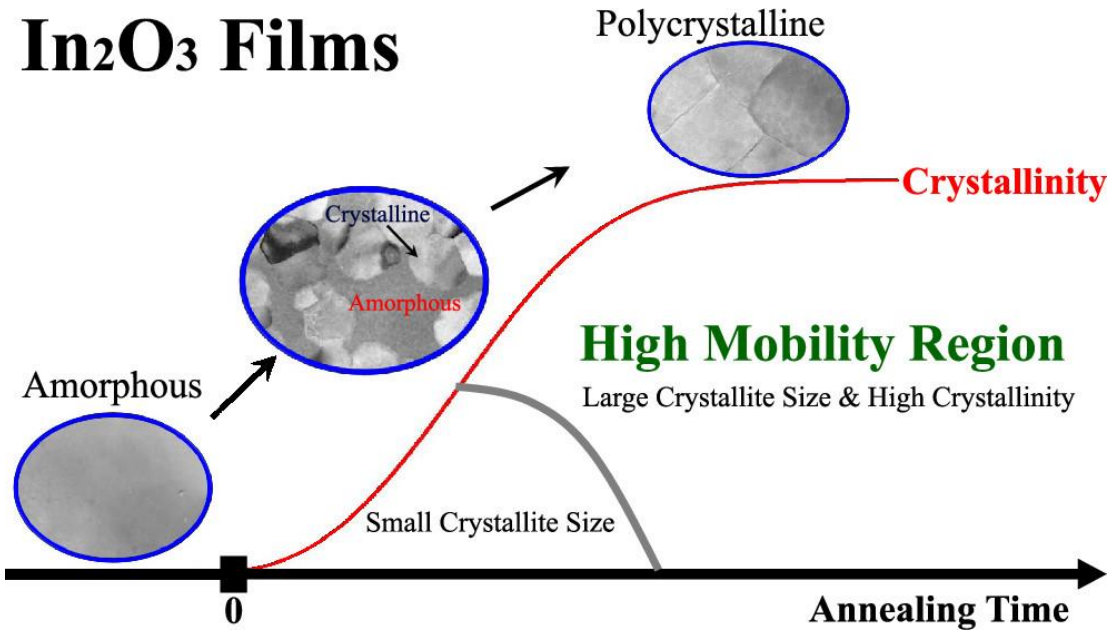
The need to understand the crystallization mechanism of the indium-based oxide films

Correlation between crystallization and device performance

- Amorphous structures and the generation of nano- to microsized crystallites with preferred crystallographic orientations have been discussed to improve the stability and performance of amorphous IGZO TFT devices

The need to understand the crystallization mechanism of oxide thin films

- Understand the crystallization mechanism and related properties of amorphous oxide thin films,
  - Annealing condition design,
  - New design insights can be obtained for preventing performance degradation of flexible electronic devices.
- Therefore, experimental approaches are needed to study important phenomena related to crystallization.
  (particularly the temporal evolution of electrical properties)

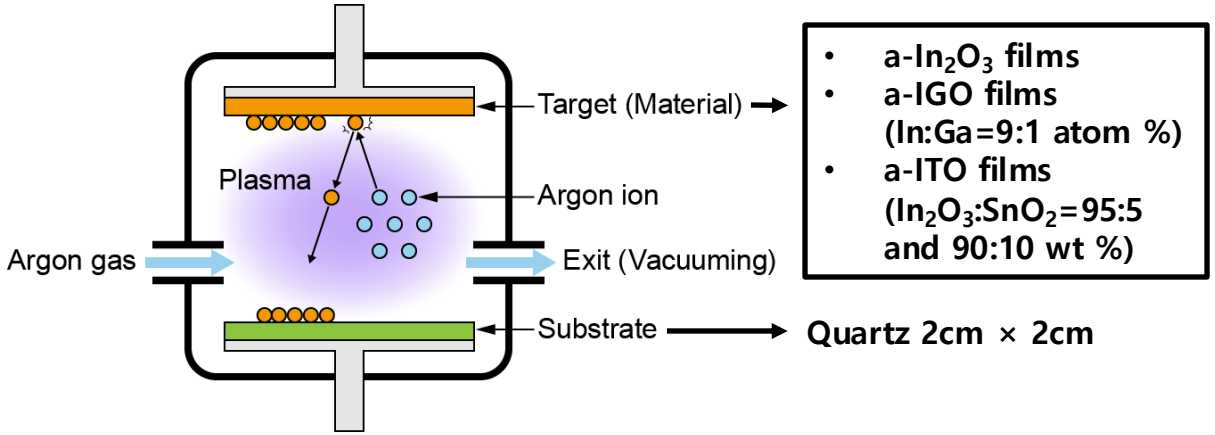


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Experimental details

Experimental investigation of the crystallization of amorphous indium-based oxide films

DC magnetron sputtering

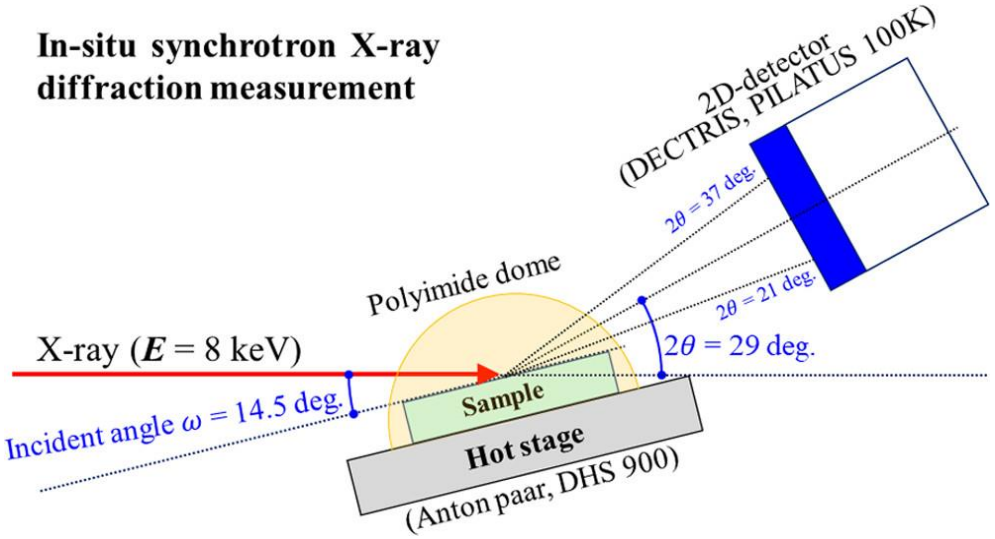


- The samples were annealed in air to obtain the temporal evolution of electrical properties
- All films are ~200nm thicks

High resolution transmission electron microscope (HRTEM)

- To analyze the formation of nanocrystallites and the crystal growth behavior after annealing

In-situ synchrotron X-ray diffraction(XRD)



- To monitor the crystallization behavior of the amorphous films throughout the annealing process
- In_2O_3 and IGO films were annealed from 150 up to 730°C in air on the sample stage during the XRD measurements

[Sputtering | Power Supply Application | Matsusada Precision]

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Structural evolution determined by XRD measurement

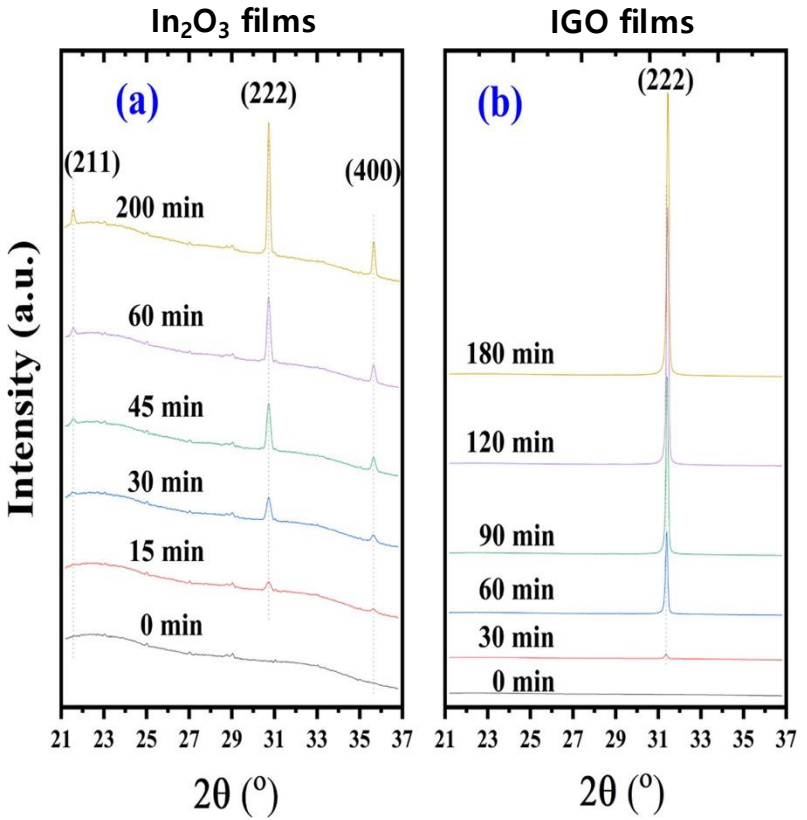
The XRD diffraction peak of In_2O_3-based amorphous films during annealing

The XRD pattern of the In_2O_3 films

- The three diffraction peaks in the XRD pattern are attributable to (211), (222), (400) faces of the In_2O_3 film
  → poly crystalline nature of the annealed In_2O_3 film
- Same peaks were observed in the XRD pattern annealed a-ITO films
  → Annealed In_2O_3-based oxides tend to transform into a bixbyite crystal structure

The XRD pattern of the IGO(In:Ga=9:1 atom %) films

- Only the (222) diffraction peak emerged in XRD pattern of the a-IGO film
  1) Partial crystallization
  2) Preferred orientation
  3) Crystal growth suppression by Ga



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Structural evolution determined by XRD measurement

The diffraction peak shift of In_2O_3 and IGO films

After annealing of 180 minute(a-In_2O_3, a-IGO films)

- The (222) peak shift
  - In_2O_3 film: ~30.7°
  - a-IGO film: ~31.4°
- The XRD peak shifted to the right(diffraction angle increased) → The reduction of lattice spacing

The different ionic radii of In^{3+} and Ga^{3+}

- In^{3+}: 0.094nm, Ga^{3+}: 0.076nm(more smaller than In)
- The peak shift toward the higher angle in the annealed a-IGO implies a decrease in the lattice spacing, most likely caused by substitution of Ga atoms into the host In sites during the crystallization

  → Ga doping affected the crystal structure formation

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Structural evolution determined by XRD measurement

In^{3+} substitution tendency for Ga site according to ion size and repulsion

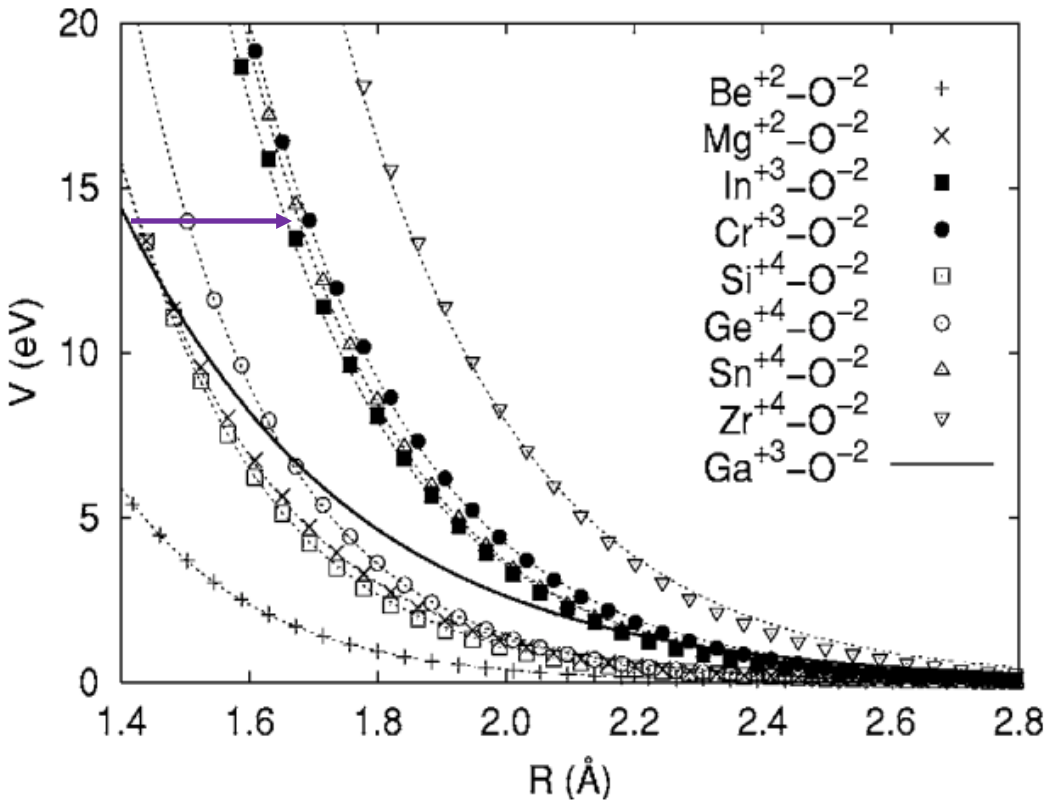
Ga-based oxide structures(Ga_I, Ga_{II})

- Within the crystal structure, Ga exists in two locations
  - Ga_I: low coordination number → less oxygen → small space
  - Ga_{II}: high coordination number → more oxygen → large space

Factors determining the position preference of ions

1. Ion size → Need more space?
2. Repulsion with oxygen ions(short-range repulsion)

- In^{3+} is a heteroelement that can replace Ga^{3+} with a similar or slightly larger ionic radius and the same charge (3+).
- The graph moves to the right ⇒ R ↑ ⇒ far from oxygen ⇒ large ion
- In^{3+} is larger than Ga^{3+} and requires more space.



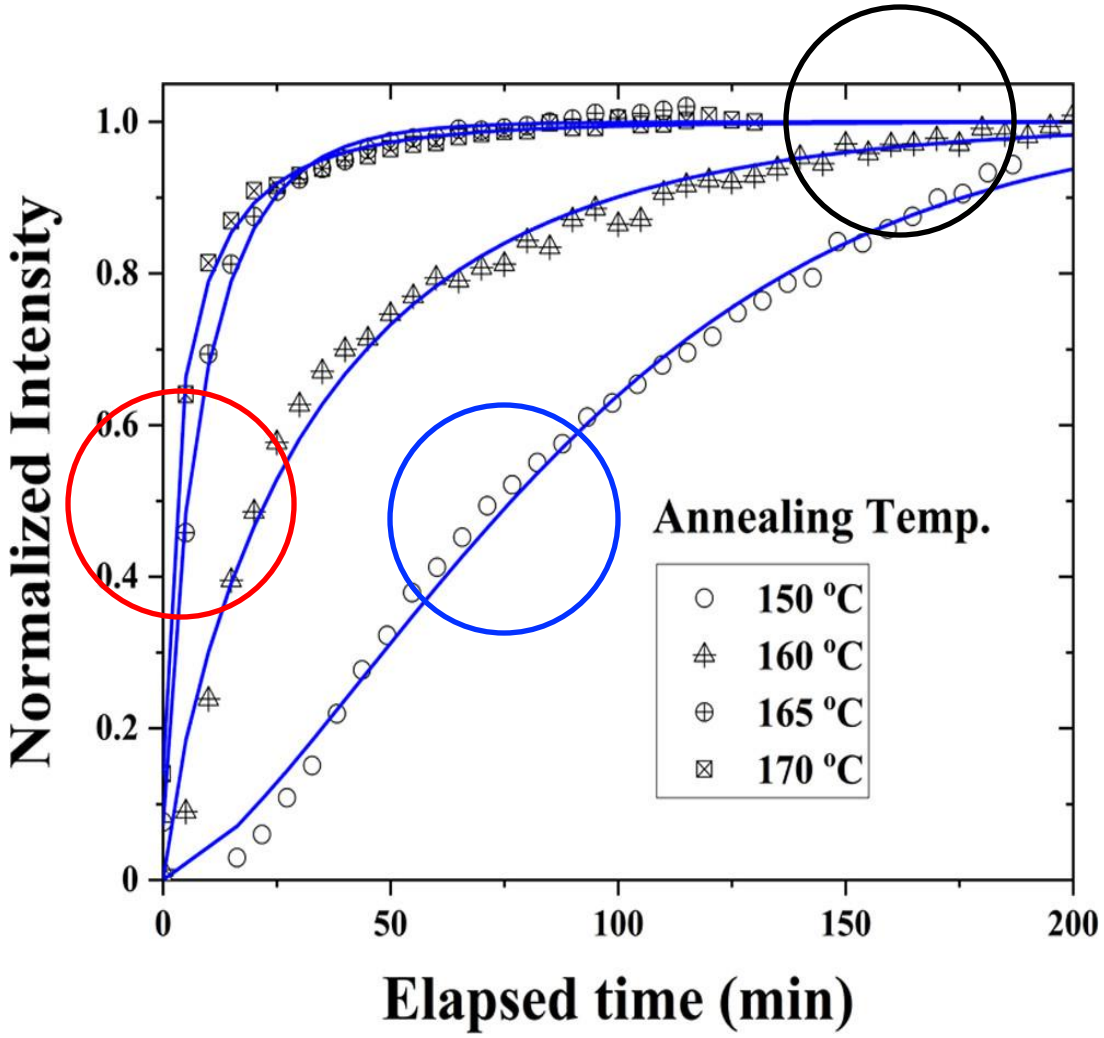
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Structural evolution determined by XRD measurement

Crystallization behavior of In_2O_3 films

Three regions of crystallization curve

1. **Initial region(Incubation time)**
   - The stage before crystal structures are formed.
   - Step of accumulating activation energy of nucleation
   - Incubation time is prominent at 150℃
2. **Middle region(Rapid crystallization)**
   - After nucleation, the crystal then grows rapidly
   - Steep slope → fast crystallization
3. Saturation region(Saturation)
   - The film slowly approaches full crystallinity
   - Stabilization zone before complete crystallization



Effects of Increasing annealing temperature

- High temperature → Rapid crystallization process(especially nucleation)
- Low temperature → Long incubation time, slow crystallization process

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Crystallization kinetics

How can we model the temporal progression of crystallization?

JMAK(Johnson-Mehl-Avrami-Kolmogorov) equation

- A model that expresses how much of the total volume has been crystallized over time.

$$f(t) = 1 - exp(-K(t-\tau)^n)$$

  - $f(t)$: $Fractional\ amout\ of\ crystallization\ as\ a\ function\ of\ time$
  - τ : $Incubation\ time\ before\ the\ start\ of\ nucleation$
  - K : $constant\ connected\ with\ the\ crystallization\ rate\ constant\ of\ isothermal\ crystallization$
  - n : Avrami exponent

- The K value changes depending on the temperature

$$K = K_0\ exp(-\frac{E_a}{kT})$$

  - E_a : $Activation\ energy\ for\ crystallization$
  - K_0 : $prefactor(constant)$
  - k : $Boltzmann\ constant$

- The higher the temperature → the larger the K value and the faster the crystallization rate
- The blue line curve(page 8.) is modeling actual measurement data using the JMAK equation

  → JMAK model explains the actual experimental results well

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Crystallization kinetics

Evaluation of $t_{1/2}$ and Avrami exponent in isothermal crystallization of In_2O_3

The crystallization half-time $t_{1/2}$

$$t_{1/2} = \left(\frac{ln2}{K}\right)^{1/n}$$

- The crystallization half-time is defined the time taken for 50% crystallization to proceed
- The larger the K value, the faster the crystallization rate, the smaller $t_{1/2}$

The Avrami exponent n

- The Avrami exponent n value is indicator of the crystallization mechanism(cubic growth, planar growth, etc.)
- $n \uparrow \Rightarrow$ Complex crystallization mechanism in which nucleation and growth occur simultaneously

  $n \downarrow \Rightarrow$ Simple crystallization mechanism
- n decreased with increasing annealing temperature

  → Simplification of the crystallization mechanism(n decreasing), faster crystal growth(Annealing temperature increasing)

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Activation energy

Evaluate activation energy to understand the crystallization process

The activation energy E_a

- Using Arrhenius equation based on constant K and the crystallization half-time with $n = 1$

$$\frac{1}{t_{1/2}} = \frac{K_0}{ln2} exp(-\frac{E_a}{kT})$$

  - Measuring $t_{1/2}$ values at various temperatures, $ln(t_{1/2})$ vs $1/T$ fitting → extract E_a from slope

Extended equation: Calculating a local E_a for a given crystallized fraction x

$$\frac{1}{t_x} = \frac{K_0}{ln(x)} exp(-\frac{E_a}{kT})$$

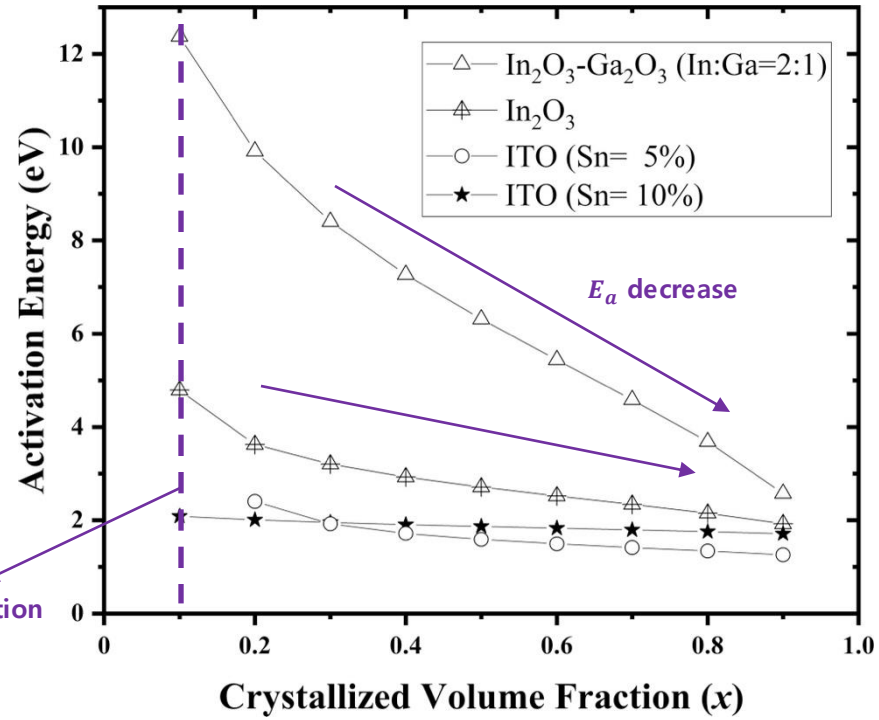
- Using this equation, we can find the energy required at any point in crystallization
- It is possible to analyze how E_a changes as crystallization progresses

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Activation energy

The relation between the activation energy and crystallized volume fraction of a-In_2O_3 films.

Evolution of the local effective activation energy with the crystalline fraction of In_2O_3 films



Initial E_a is large
→ More energy is needed in the initial stages of nucleation

- As crystallization progresses to a certain level, less energy is required during the crystal growth stage → E_a decrease
- Crystallization is most difficult at the beginning and becomes easier as it progresses.
- For a-In_2O_3, the E_a value at $t_{1/2}$ is estimated as 2.71eV → A reference value that can be compared with other materials

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Dopant effects

The effect of impurity elements on the activation energy(in the early stage)

The addition of the Ga element to amorphous In_2O_3(a-IGO)

- The local activation energy in the early stage of crystallization increases → The energy barrier to crystallization is increased
- Nucleation is difficult and crystallization requires a high annealing temperature and long $t_{1/2}$
- Impurities generally tend to be excluded from homogeneous nucleation

The addition of the Sn element to amorphous In_2O_3(a-ITO)

- local E_a decreased → more easier crystallization
- Increasing Sn concentration accelerates the crystal growth in the early stage
  - the local E_a in ITO(Sn 5%) is larger than that of ITO(Sn 10%) in the early stage(x < 0.3, in the early stage)

⇒ Crystallization characteristics can vary significantly depending on the type and concentration of dopant

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Dopant effects

The effect of impurity elements on the activation energy(in the late stage)

The late stage of crystallization(x > 0.5)

- More than half of the overall crystallization has progressed

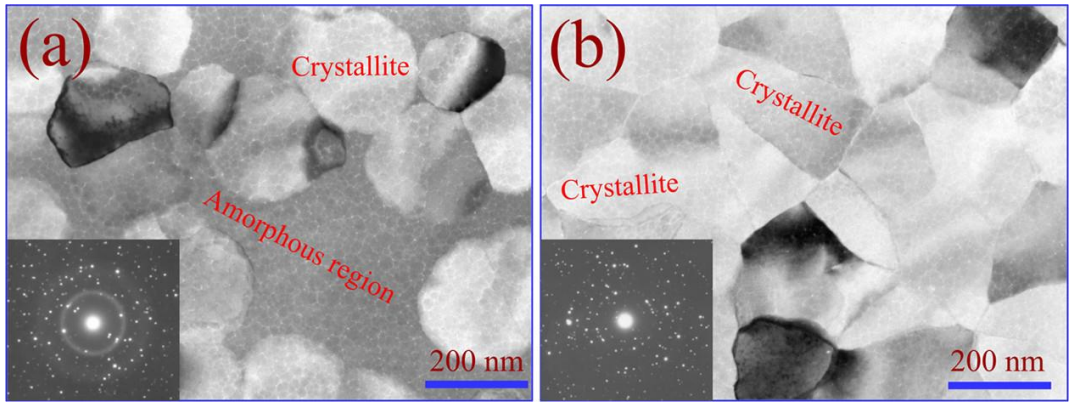
  → The process in which most of the nuclei are formed and the remaining amorphous parts grow to complete crystallization
- In the late stage of crystallization, the speed at which atoms move and the crystal grows determines the overall speed

Diffusion-controlled growth

- For a crystal to grow, surrounding atoms must move into the crystal structure
  - In indium-based oxides, The most mobile atom during crystal growth is the oxygen atom
  - Cation(In^{3+}) vacancy diffusion in In_2O_3 was negligibly low
- In the late stage, the local E_a was lower in ITO(5%) than in ITO(10%)

  → Crystallization occurs easily and quickly in ITO(5%)
  - Sn elements promote oxygen diffusion(migration)
  - In_2O_3 + Sn^{4+} ⇒ Oxygen vacancies ↑ ⇒ Diffusion path!!

⇒ The diffusion rates of oxygen determine the rate of crystal growth



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Dopant effects

Defect types and oxygen diffusion mechanisms

Diffusion Mechanism in In_2O_3 and IGO: Vacancy mechanism

- Oxygen vacancies are determined as the predominant defect type in In_2O_3
- Ga-O bond is stronger than In-O bond

  → Oxygen deficiency around the indium ion is considered as the predominant defect type in a-IGO
- Oxygen diffusion would take place via a vacancy mechanism → Oxygen atoms move through the vacancy

Diffusion mechanism in ITO: Interstitial mechanism

- For a-ITO films(5%), the carrier density increases from 6.29×10^{19} to 5.20×10^{20} cm^{-3} after annealing at 200°C for 1h

  → Sn atoms are substitutionally doped into In_2O_3 and become electron donors
- When Sn^{4+} is added, oxygen interstitial is generated to balance the charge.
- As a result, oxygen moves in the ITO film by diffusing through the interstitial spaces

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Dopant effects

Oxygen interstitial due to Sn^{4+} doping charge compensation

Increasing electron concentration by Sn doping

- In In_2O_3, the In atom is a trivalent ion(In^{3+}), and doping Sn here means substituting Sn^{4+} in the In^{3+} position.
- When Sn^{4+} enters, it exists in a state where it releases one more electron than In^{3+}.

  → From the perspective of the entire lattice, there is one excess electron(Increased electrical conductivity)

⇒ Sn^{4+} acts as a donor, supplying one more electron than In^{3+}

Interstitial oxygen production by charge compensation

- There must always be a charge neutrality within the lattice. → Additional positive charge due to Sn^{4+}?
- Oxygen ions with negative charge(O^{2-}) enter the interstitial site, thereby compensating for the overall charge balance.

  → Oxygen interstitial(O_i^{2-}) increases within the lattice

⇒ Oxygen diffusion is considered to be dominated by interstitial migration mechanism in crystallized ITO films in the late stage of crystallization

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Dopant effects

Oxygen diffusion Energy Comparison(Experiment vs Theory)

Calculated migration energy in crystalline In_2O_3

defect	Charge state	Migration energy(eV)
Oxygen vacancy(V_O)	+2	1.10 eV
Oxygen interstitial(O_i)	-2	1.40 eV

Experimentally extrapolated E_a (at x = 0.9)

Films	E_a	mechanism
In_2O_3	1.85 eV	Vacancy, slow
IGO	2.57 eV	Vacancy, very slow
ITO(5%)	1.26 eV	Interstitial, fast

Comparison Experimental and calculated E_a

- In ITO(5%) films, experimental E_a ≈ theoretical values
  → Interstitial mechanism is correct!
- In In_2O_3 & IGO films, experimental E_a > theoretical value
  → Low V_O concentration(Diffusion was less or more difficult)

Actual oxygen vacancies and oxygen interstitial concentration

- Very low oxygen vacancy concentration → Difficulty oxygen vacancy diffusion
- ITO(5%) films have a lot of oxygen interstitial. → Diffusion proceeds easily through interstitial mechanisms.

Amorphous structure affects diffusion mechanism

- The amorphous state does not have a complete bixbyite structure.
  → For this reason, the actual diffusion barrier(energy) also differs from the theoretical value.
- There are spaces in the amorphous structure where oxygen interstitial can be located.

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Dopant effects

Relationship between Sn doping and increased electron concentration

Slow electron concentration increase in ITO(10%)

- In ITO(5%) films
  - Electron concentration after 200 ℃ annealing: $6.29 \times 10^{19} \rightarrow 5.20 \times 10^{20}$ cm^{-3}
- In ITO(10%) films
  - Electron concentration after 300 ℃ annealing: $6.29 \times 10^{19} \rightarrow 2.75 \times 10^{20}$ cm^{-3}

⇒ Even though Sn increases, the electron concentration becomes saturated or increases more slowly.

Formation of an inactive structure, SnO_8

- Sn concentration 5% ↑ ⇒ Sn atoms form the SnO_8 structure, instead of replacing the In site.
- SnO_8 becomes electrically inactive, which means it cannot supply electrons, so the doping efficiency decreases sharply.
- The SnO_8 structure is strongly bound to O^{2-}, making it difficult for oxygen ions to move interstitially.

  → Decreased oxygen interstitial concentration(Reduced oxygen diffusion rate)

  → Leading the higher E_a

⇒ Therefore, ITO(10%) is more difficult to crystallize and progresses slowly.

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Growth morphology

Component of Avrami exponent equation

Avrami exponent equation

$$n = b + pm$$

- b : $Nucleation\ index$
- p : Growth index
- m : $Growth\ dimensionality$

- b (Nucleation index)
  - $b = 0$: Nucleation occurs only at the beginning.
  - $b = 1$: New nuclei are added to a system at a constant nucleation rate.
  - $b > 1$: The nucleation rate grows.
  - $0 < b < 1$: The nucleation rate decreases over time.

- p (Growth index)
  - $p = 1$: Interface-controlled growth
  - $p = 0.5$: Diffusion-controlled growth

- m (growth dimensionality)
  - $m = 1$: 1-dimentional growth
  - $m = 2$: 2-dimentional growth
  - $m = 3$: 3-dimentional growth

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Growth morphology

Limitations of JMAK Analysis

Temperature dependence of n values

- At 150 ℃, $n \approx 2$ & At 170 ℃, $n \approx 0.5$

  → The crystallization mode changed from 3D to 1D as the annealing temperature increased. (???)

- Generally, with high annealing temperature

  → Thermal energy is sufficient to overcome the energy barrier

  → Initiate more nucleation

  → Enhancing the crystal growth during the crystallization process.(Increasing n value)

⇒ The JMAK equation alone has limitations in accurately interpreting crystallization behavior.

Three assumptions of JMAK equation

1) The system is infinitely larger than the individual growing particles.
2) The nucleation is randomly homogeneous (bulk nucleation) throughout the material.
3) The particles cease growing at points of mutual contact but continue growing elsewhere.

- But In the case of thin film crystallization, the assumptions of the JMAK equation are violated. → two reasons
  1) The growing particles impinge upon free surfaces, violating the infinite size assumption.
  2) Heterogeneous nucleation from an interface is favored in many systems, violating the random homogeneous nucleation assumption.

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Growth morphology

Limitations of JMAK equation and local Avrami exponent

Local Avrami exponent $n(x)$

$$n(x) = \frac{\partial ln(-ln[1-x(t)])}{\partial ln(t-\tau)}$$

- The correlation between the local Avrami exponent $n(x)$ and the crystallized volume fraction x
- The above formula more realistically describes the temporal evolution of crystallization.

Difficult to measure the temporal variation of $n(x)$

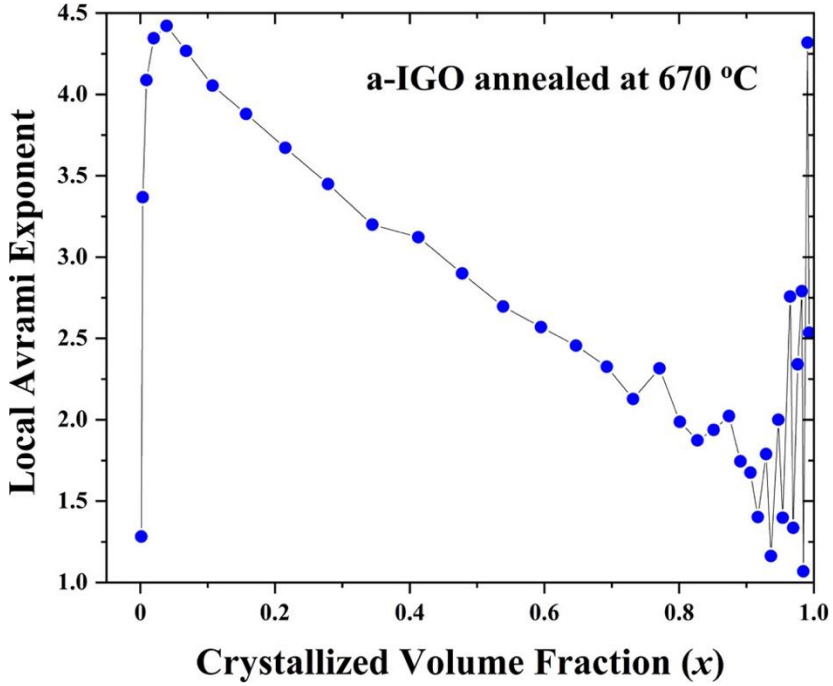
- A changing $n(x)$ is difficult to obtain because a-In_2O_3 and ITO films rapidly crystallize, as indicated by a very short $t_{1/2}$.
- A-IGO with a high activation energy was selected as a suitable material for investigating the temporal evolution of $n(x)$.
  - High activation energy → Slow crystallization rate → $n(x)$ changes can be tracked more precisely
- Local Avrami exponent $n(x)$ versus crystallized fraction x (670 ℃ annealed IGO film) derived from JMAK equation.

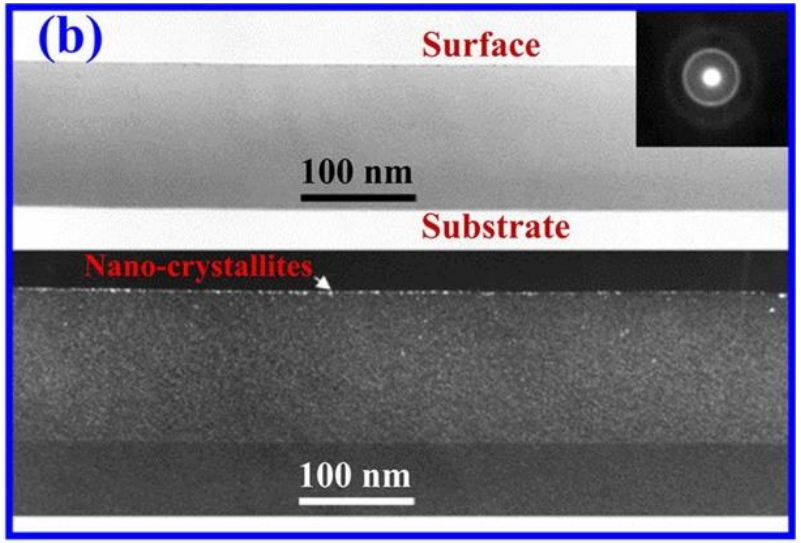
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Growth morphology

$n(x)$ in the early stage crystallization

Initial stage ($x \approx 0$) : Crystallization initiated at the surface





- **At the beginning of crystallization, $n(x) \approx 1.2$**

  → Constant interface-controlled crystallization growth

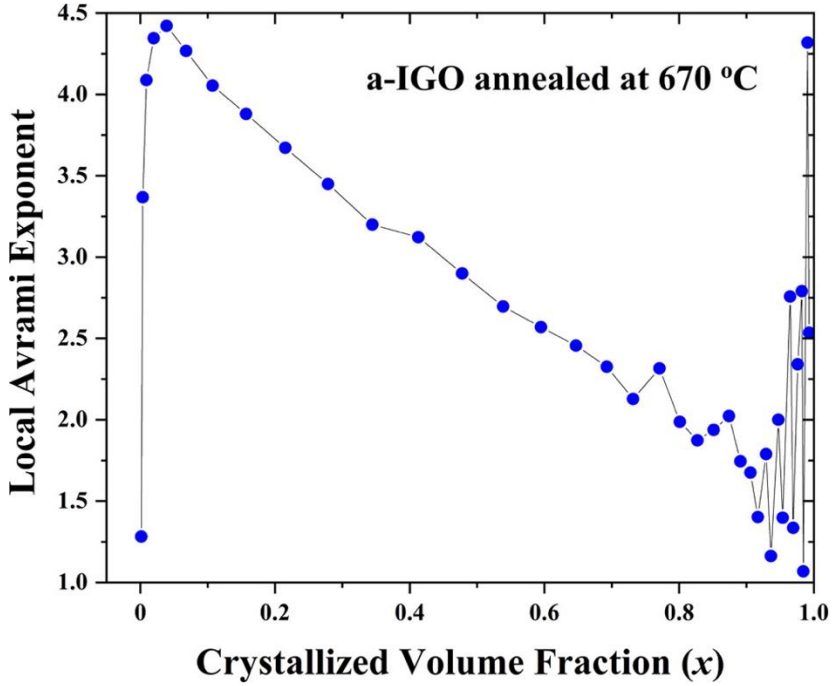
- **The crystallites emerged from the film surface.**

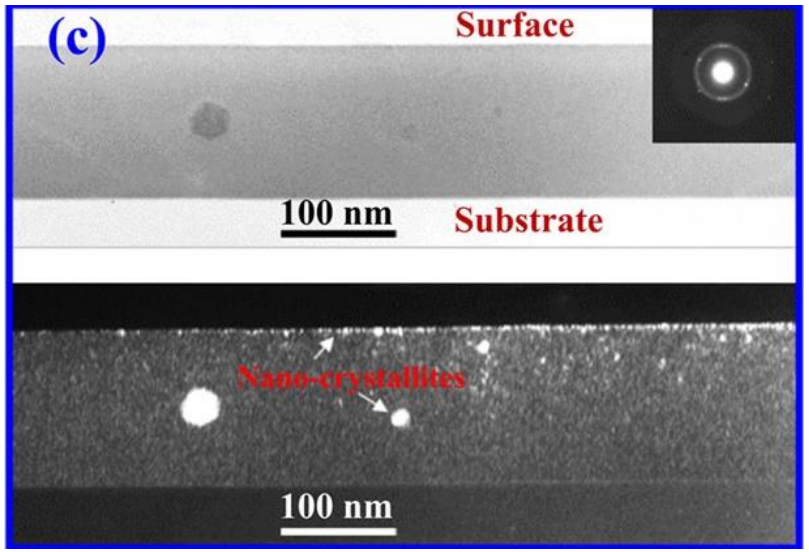
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Growth morphology

$n(x)$ in the mid stage crystallization

Middle stage ($x \approx 0.05$) : Rapid $n(x)$ increase





- As the crystallization continued, $n(x)$ rapidly increased to 4.5 at x = 5%
- Growth from crystal nuclei formed on the surface, At the same time, it shows new nuclei forming and growing inside the film.

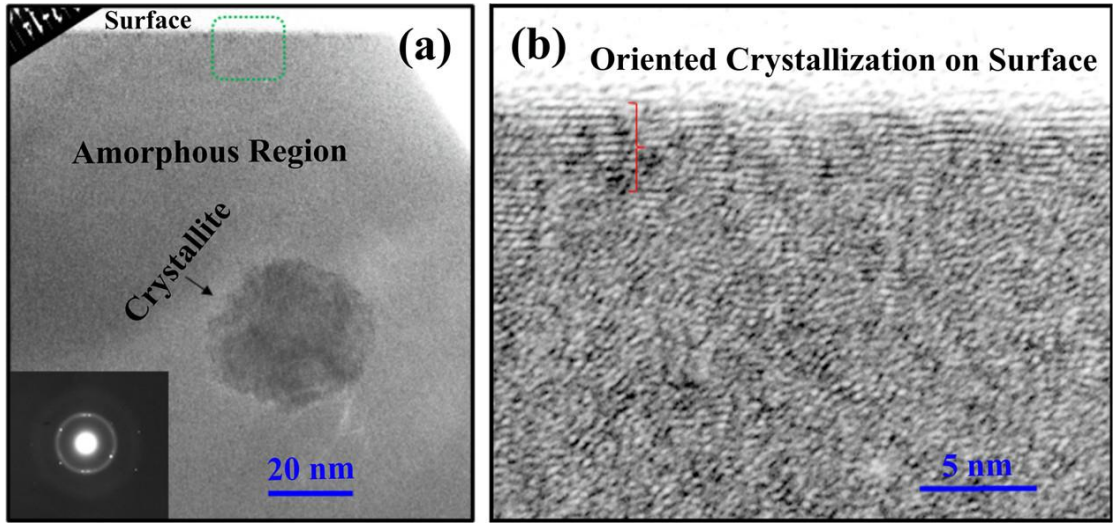
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Growth morphology

$n(x)$ in the mid stage crystallization

Surface growth and internal growth of crystal nuclei

- The crystals now grew from pre-existing surface nuclei while simultaneous nucleation and growth occurred inside the film.



Cause of increase n(x) ≈ 4.5

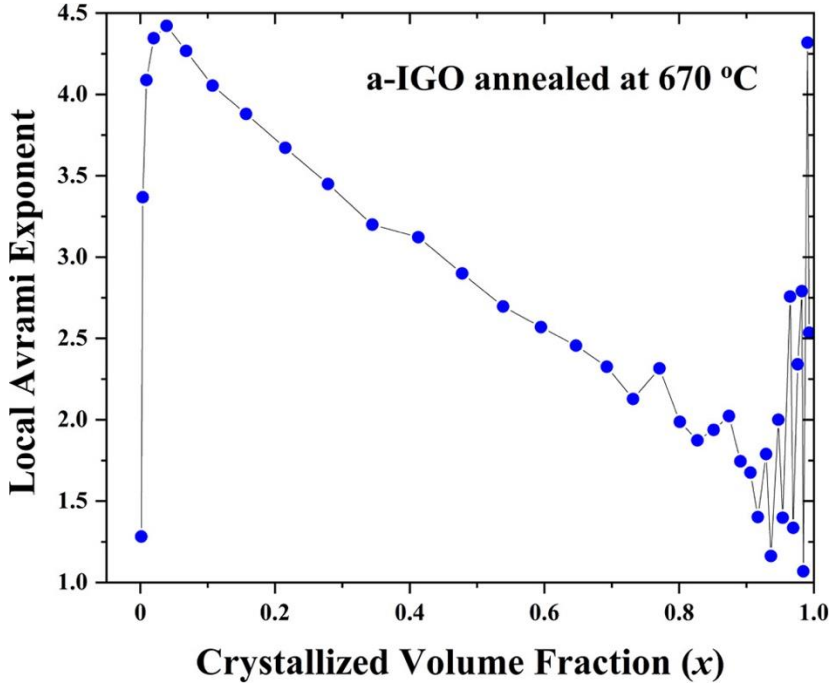
- The nanocrystallites were very much smaller than the film thickness (m=3)
- A growing nucleation rate (b > 1) is expected in the early stage of crystallization.
- $n(x) = m * b = 3 * 1.5 \approx 4.5$

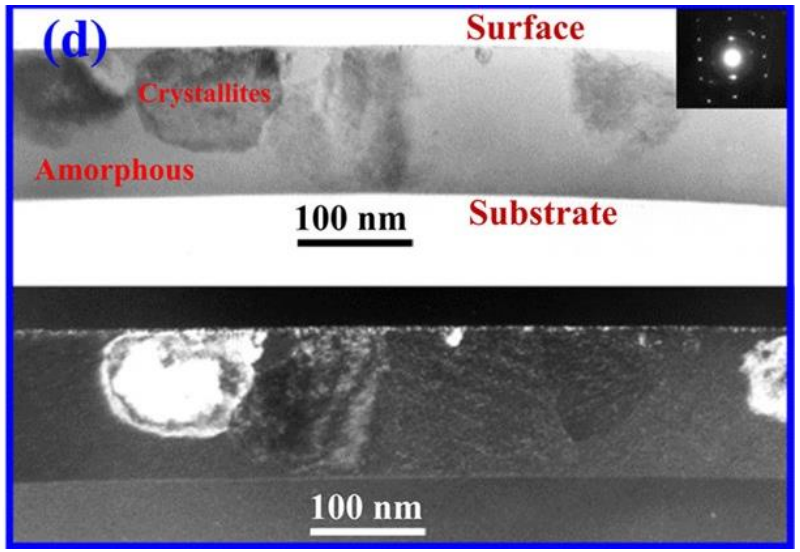
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Growth morphology

$n(x)$ in the later stage crystallization

Late stage ($x \approx 0.90$) : Planar growth & diffusion control due to thin thickness





- When the growing crystallites reached the film surface, they were confined to lateral growth by the finite film thickness, and the growth mode becomes two-dimensional.
- The amorphous volume decreased and the crystal growth became diffusion-controlled in the late stage.

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Film Crystallinity and Electrical Properties

Carrier concentration change trend after annealing

Carrier concentration reduction

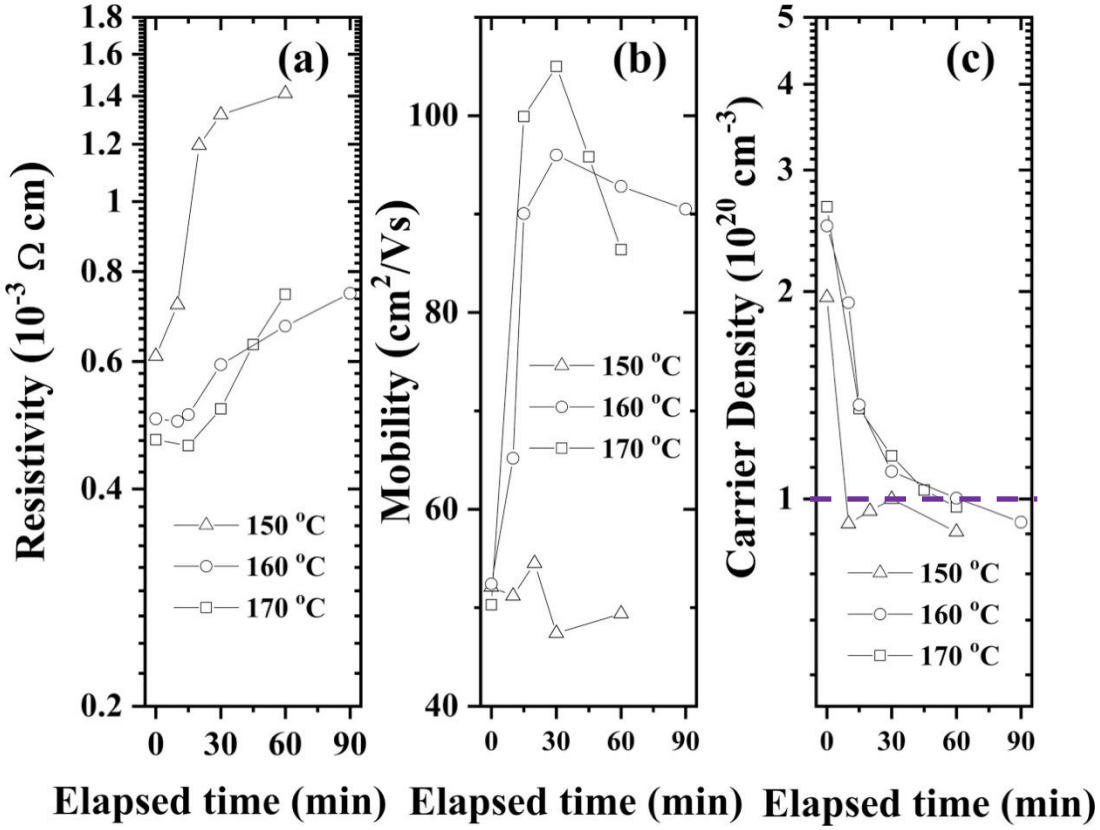
- At annealing times of ~30 min or longer, the carrier density decreased to a lower plateau(~1 × 10^{20} cm^{-3}) regardless of the annealing temperature.

Causes of decreased carrier concentration : V_O reduction

- Positively charged oxygen vacancies(V_O) are the predominant defect type in amorphous indium-based oxide films.
  →Oxygen vacancy creates free electrons and increases conductivity, and when this vacancy decreases, conductivity also decreases.

Changes in oxygen vacancy due to oxygen diffusion

- After annealing in air for 30 min, the oxygen diffuses into the films and fills the vacancies.
- V_O ↓ ⇒ Free electrons ↓ ⇒ Carrier concentration ↓
- A long annealing duration did not obviously decrease the carrier density. → Saturation



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Film Crystallinity and Electrical Properties

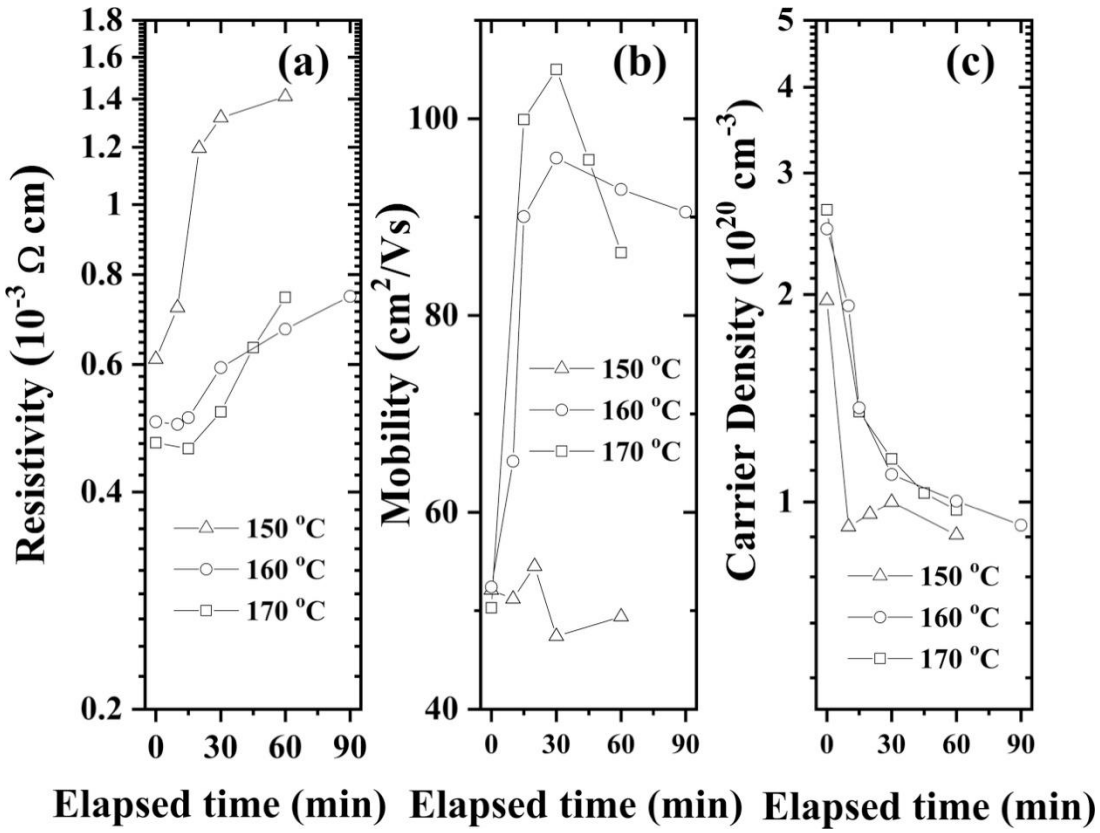
Mobility change trend after annealing

Changes in mobility depending on temperature and annealing time

- The maximal mobility exceeds 100 $cm^2/V{\cdot}s$ at 170 °C for In2O3 films annealed for 30 min.
- At 160 and 170℃ annealing, annealing time ↑ ⇒ mobility ↓
  - Excessive annealing can cause structural defects or oxygen re-diffusion, which can hinder electron transfer.
- At 150 ℃ annealing, the mobility remained low and independent of the annealing time.

Correlation between degree of crystallization and mobility

- Based on the same 30-minute annealing time,
  - 150℃ fraction of the crystallized volume : 16.4% (amorphous)
  - 160℃ and 170℃ : 62.7% and 92.9% (more crystallization)
- A smaller mobility was observed in those films with smaller crystallite sizes.



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Film Crystallinity and Electrical Properties

Mobility changes related to crystallization

Cause of low mobility : Amorphous barrier between crystals

- Small mobilities at 150°C can be attributed to the existence of amorphous phases between crystallites, acting as an energy barrier for the electron conduction.
- Single isolated grains crystallize in the amorphous phase below the percolation threshold (20% volume fraction).

Reason for high mobility : Linked grain structure

- A mobility of > 90 $cm^2/V{\cdot}s$ can be implemented for In_2O_3 films with a crystallinity of > 40% and a crystallite size of > 70nm.
- When this condition(>40%, >70%) is met, the percolation threshold is crossed, and a continuous path is formed through which electrons can move between grains.

Influence of grain boundaries and low barriers

- Grain boundaries can interrupt electron movement, and when the charge concentration is ~10^{20} cm^{-3}, this barrier height is reduced to about 25meV.

  → This small grain boundary barrier is considered to contribute to the improved mobility.

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